

maintain absolute concentration through long, exhausting shifts, fixed in front of their screens. The controllers had to interpret data from different sources – including television cameras carried by RPVs over enemy missile sites – and turn this rich and often confusing flood of information into a coherent picture of the air battlefield.

Either through a data link to their onboard computers or by voice over UHF radio, the AWACS controllers kept pilots informed of developments on a minute-by-minute basis. In effect, the controllers could orchestrate the whole battle, giving coherence to the approximately 3,000 sorties a day flown by Coalition forces.

Information from AWACS often allowed Coalition fighters to engage Iraqi aircraft beyond visual range – some Iraqi airmen were downed by missiles fired by aircraft they never saw and of whose presence they were totally unaware. The Iraqis' aircraft shelters proved vulnerable to American precision weapons when their airfields came under attack. The Iraqi air force soon admitted its impotence and gave up the fight, seeking sanctuary in Iran. Iraqi air space was mostly penetrated by the same basic mix of aircraft that had been employed for such missions in North Vietnam, but operating with much improved technology. There might be EF-111s specialized in electronic countermeasures jamming Iraqi radars; F-4s armed with anti-radiation missiles flying Wild Weasel missions to destroy Iraqi SAM sites; a core of strike aircraft for the ground-attack mission; and mostly F-15s escorting the "package" on combat air patrol.

### Stealth technology

One aircraft, however, needed no help from others in penetrating Iraqi defenses. The F-117 radar-invisible stealth aircraft was used in combat in the Gulf War and resoundingly justified the amount that had been spent on it. Unafraid of radar, it did not have to fly at the ground-hugging altitude adopted by most modern strike aircraft to penetrate air-defense systems. Nor did it need to be accompanied by escort fighters, electronic-warfare aircraft, or antiradar strike aircraft. Its pilot flew alone through the night in radio silence, eyes fixed on the screens that allowed him to navigate a path into enemy territory and to locate and illuminate his target. Usually the target was destroyed and the F-117 would return as it had come, invisible as a ghost, leaving the enemy guns to blast away blindly into the night sky in the vain hope of a lucky hit.

An aircraft's radar cross-section – the degree to which it shows up on radar – is a function of its shape. Since radar depends on signals striking an

## Lockheed F-117 Nighthawk



**IN 1975, THE LOCKHEED** Skunk Works initiated a secret program to develop a radar-proof "stealth" aircraft. Development was by no means straightforward, and the first F-117As were not delivered to the USAF until 1982. Nicknamed the "Bat Plane" or "Stink Bug," its unique appearance resulted from the use of flat, faceted surfaces to deflect radar emissions at an angle, especially from the front.

The airframe was covered in radar-absorbent material and the cockpit windows were coated in a gold layer that conducted radar energy into the airframe. There was even a fine mesh over the engine intakes, which might otherwise provide a radar signature for the enemy to track. Any weaponry or fuel tanks were carried internally to maintain the stealth effect. It could not use radar, because this might have given away its position, relying instead on a passive infrared "eye" in the

nose to look forward at the ground. The result was an aircraft as near to radar-invisible as possible. The pilot's view out of the cockpit was very limited, but this did not matter because the aircraft's subsonic speed made it too slow for daylight combat. It only attacked by night, and the pilot never looked outside, as he kept his eyes on the multi-function display screens in front of him.

The designation "F" for fighter was a strange one, as the F-117A had no guns and did not usually carry missiles. A night-penetration strike aircraft, it was equipped instead with the most up-to-date smart weapons and targeting systems for precision attack. While the F-117A was a complex weapons system to be operated by a single pilot, who needed a lot of electronic help approaching the target, its strikes proved extremely accurate in the Gulf War in 1991.

The F-117As justified their impressive price tags – over \$40 million each – by their accuracy of attack and near invulnerability, although one was shot down by the Serbian artillery during the Kosovo campaign in 1999. The aircraft was retired from service in 2008, but is in recallable storage; some flew in 2016.

*All glazed panels are gold-coated to conduct radar energy into the airframe*

*Faceted surfaces give angular appearance*

*Engine housing*

### INVISIBLE NIGHTHAWK

*Like many other weapons, the F-117A was developed to counter advances in an enemy's military capabilities. In this case, however, the F-117A not only raised the stakes, but also bypassed an entire class of weapons entirely – radar-directed air defenses.*

*Hydraulically actuated weapons launch trapeze mechanism in top of weapons bays*

